

Guided Self-Study: Odor Code Cracker

Introduction

Olfaction is the scientific term that describes our sense of smell. Smell is an incredibly important sensation. It guides our food seeking and selection, can produce arousal/attention, social selections, produces aversion, or serve as a danger signal. Despite being so important in our day to day lives, we often don't think about it until a smell catches our attention. Like many of our senses, smell, starts at the nose where the chemicals in the air, or **odorants**, are detected. Specialized neurons called olfactory receptor neurons (ORNs) convert this chemical signal to an electrical impulse that travels through the brain. This exercise combines guided reading, visual modeling, and an interactive puzzle that will help you explore how the brain encodes smells using receptors, circuits, and brain maps.

Deliverables

A separate answer sheet is provided for you on UB LEARNs. All of the questions for each activity appear here as well as on the answer sheet for your convenience. In addition to written answers, you will be asked to create a simplified image of the olfactory tract. This can be hand drawn, generated on BioRender, PowerPoint, or whatever medium you want – this is your time to get creative and think about how you would visually model this. This should not be an in-depth art-style project. This image can be inserted into the answer sheet or uploaded separately on UB LEARNs.

Note, the figures that appear in this document were created in BioRender or adapted from Neuroscience by Purves, Dale.

Learning Objectives:

This activity is designed to prepare you to:

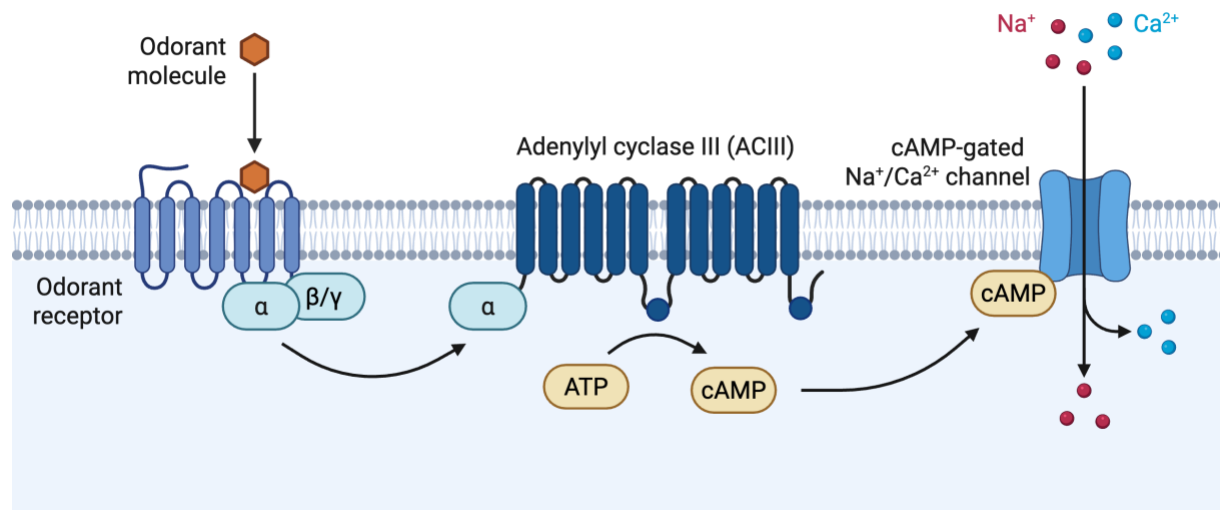
1. **Describe** the signal transduction pathway in olfactory neurons (ORNs).
2. **Trace** the neuronal pathway of olfactory information from the nasal epithelium to the cortex.
3. **Interpret** how combinatorial receptor coding enables odorant discrimination.

Part I. Signal Transduction

The lectures this week all place emphasis on signal transduction – the method by which physical stimuli in the world are transformed into neuronal impulses. Smell is no

different. The air we breathe is filled with small chemical particles – some organic some inorganic. These smells can signal the availability of food, the presence of a specific person, chemicals, or danger.

Just like tastants in our food, odorants in the air interact with specific receptors called **olfactory receptors**. There are hundreds of different olfactory receptors in the human genome. Each recognizes different odorants or combinations of odorants. Structurally, all olfactory receptors are G-protein coupled receptors (GPCRs). The extracellular segments of these receptors are specialized according to the type of odorant they bind. However, ALL olfactory receptors have the same intracellular segment and use the same G-proteins, called **G_{olf}**. The events associated with G_{olf} activation and signaling are shown in the image below.



Created in BioRender for NRS 301

Part I. Think it through

Consider the image. Arrange the following events in the correct sequence (1-6)

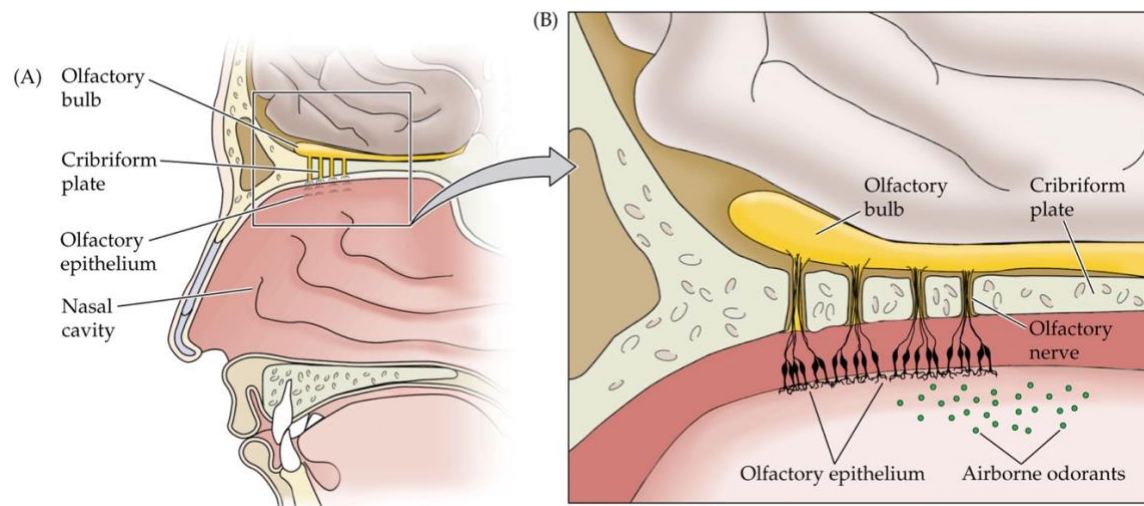
1. Na⁺/Ca²⁺ channels open, leading to depolarization
2. cAMP levels increase inside the olfactory receptor neuron (ORN)
3. Odorant binds to specific olfactory receptor
4. Olfactory G-protein (G_{olf}) is activated
5. Action potential is generated and travels along the axon
6. Adenylyl cyclase is activated and converts ATP to cAMP

What do you think would happen to your sense of smell if the Golf protein was nonfunctional? Would the receptor still bind odorants? Would the neuron still generate an action potential?

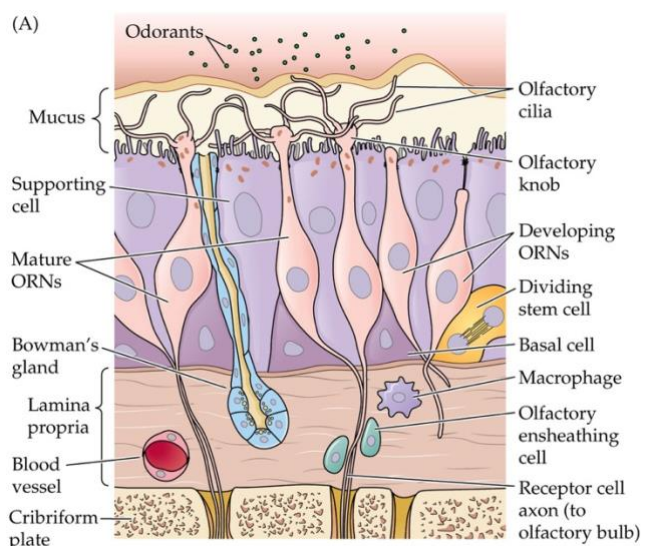
What do you think would happen to your sense of smell if adenylyl cyclase was overactive?

Part II. The Anatomy of Olfaction

Now that we know how olfactory information is detected. Let's talk about where this information travels in the brain.



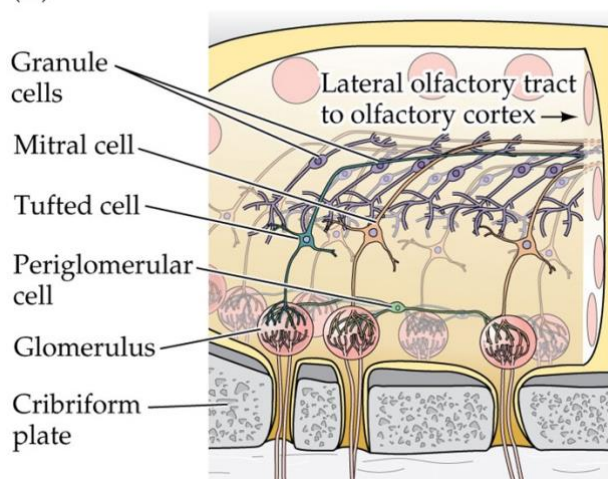
It comes as no surprise that olfaction starts at the nose. Embedded in the nasal epithelium is a set of neurons called the olfactory receptor neurons (ORNs). These neurons are named as such because it is these neurons that express the olfactory receptors and are depolarized in the presence of odorants. The position of ORNs is a



little problematic. The receptive or dendritic end of the neurons extend into the nasal cavity. This is required for the receptors to be able to interact with the odorants in the air. But this also means that these neurons are subject to damage causing ORNs to break down. However, we don't lose our sense of smell because these cells are routinely regenerated. Another notable feature about ORNs is that they are unmyelinated. Although these neurons are unmyelinated, their signaling is not impaired. This is because

ORN axons are incredibly thin and are “short” in comparison to other neurons in the nervous system.

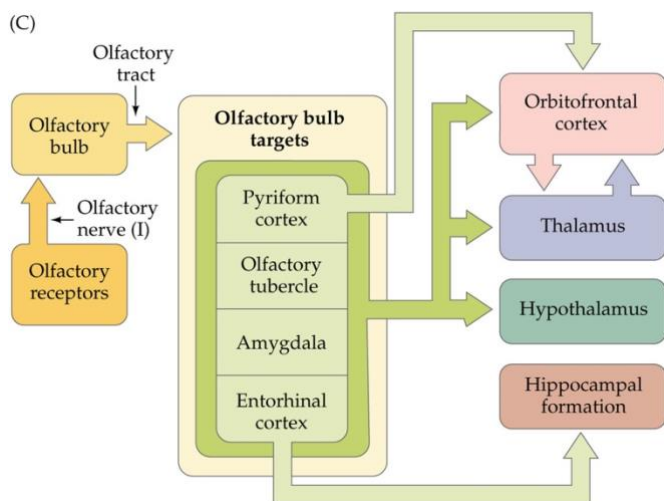
(D)



The axons of ORNs extend through the skull and immediately end in a structure known as the olfactory bulb. The olfactory bulb contains a series of neurons. Axons from ORNs synapse in a specialized structure known as a **glomerulus**. Within each glomerulus, the ORN axons connect with the dendrites of two different types of neurons, mitral and tufted cells. These cells carry the signal deeper into the brain. I think of glomeruli as synaptic bubbles – axons from ORNs meet dendrites from olfactory bulb neurons (aka tufted & mitral cells) and are contained in a

membranous sphere. This membranous sphere keeps neurotransmitter from spreading to nearby structures and isolates the neuronal signaling. This increases specificity of our signal. It is important to note that many ORNs will synapse on a single glomerulus. In this way, the glomerulus collects information from across the nasal epithelium. These glomeruli also serve as major relay stations to many other brain areas.

(C)



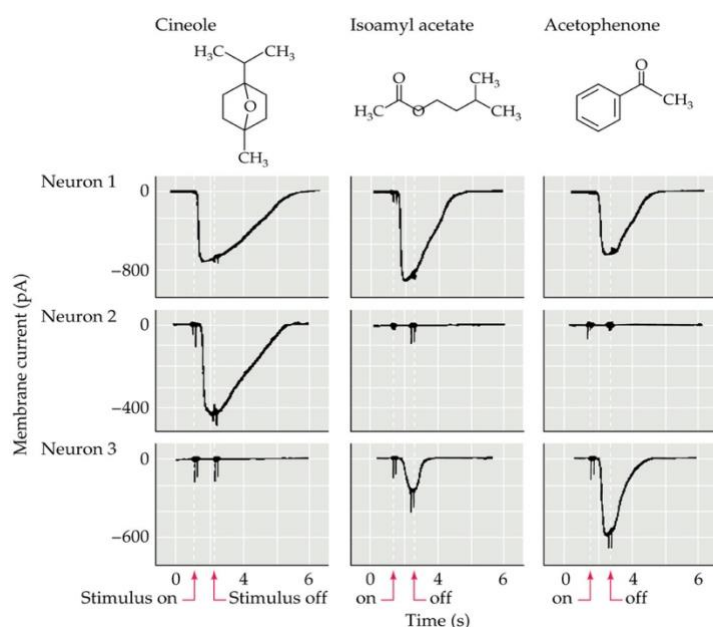
You can see how many brain areas receive olfactory information in the figure to the left. Some notable regions include the **pyriform cortex**. The pyriform cortex is the brain area on the temporal lobe that helps us determine an odorants identity (what is this smell) and its intensity. The **amygdala** helps to provide emotional context to our smells. Consider, is there a smell that reminds you of home or a place you love? Is there a smell that can help you feel calm? What about scared or

anxious? Those feelings are evoked by the amygdala. Our ability to remember those smells and their meaning to us comes from the involvement of the **hippocampus**. The **hypothalamus** works to coordinate hormonal responses to these scents. These hormones could relate to digestion (in the case of food smells) or stress (in the case of dangerous smells). The release of hormones helps to coordinate our behavioral and physiological responses to these smells.

Part II. Activity

Draw yourself a simplified schematic of the olfactory system. This should look something like the somatosensory pathways I provided for you during exam 3 (refer to the Exam 3 Cheat Sheet on UB LEARNs). In your image, you should have 6 neurons. There should be one neuron that correspond to the olfactory receptor neurons (ORNs), glomeruli, and cortical projections discussed in the paragraph above. Label each neuron and provide 2-3 key details about the neuron or the function of the brain area in olfaction.

Part III. Topographical Mapping in Olfaction

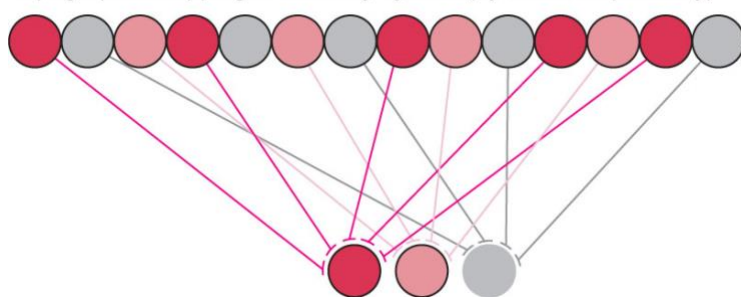


As mentioned above, each olfactory receptor will respond to a specific odorant or odorants. An example of this is shown below. Here we are considering three different odorants Cineole (eucalyptus), isoamyl acetate (banana), and acetophenone (almond/orange). The image below also shows the activity of three different neurons. If a neuron responds to an odorant, then the membrane current sharply decreases. This is evidence of depolarization in the neuron. Notice that neuron 1 responds to all three odorants. Neuron 2 only responds to

the odorant in eucalyptus. Neuron 3 responds some to the odorant found in banana but strongly responds to the odorant found in almonds and oranges.

Although ORNs for different odorants are scattered throughout the nasal epithelium, their axons converge in organized clusters (glomeruli) in the olfactory bulb. This creates a topographical map of odor information. This starts at the level of the nose and is seen at every anatomical stop in the brain. In the image below you can see an example of how this topographic mapping looks as it travels from the olfactory receptor neurons (ORNs) in the nose to the glomeruli in the olfactory bulb. Notice specifically the coloration and its organization. At the level of the nose, there is some chaos – neurons

Topographic mapping in olfactory system (by odorant specificity)



encoding different odorants are found throughout the whole nasal epithelium. At the level of the glomeruli, we start to see clustering of neurons according to the odorants they respond to. This is maintained throughout the brain in every region where olfactory information is received.

Part III. Activity

In the activity below, we will model this topographical mapping cvb. You will use the provided odorants, receptors, and hints to identify olfactory circuits.

The tables below provide you with information about how different olfactory neurons are connected, their sensitivities to different odorant qualities, and a list of odorants and their properties.

Receptor Connected Glomeruli Cortex Zones

OR1	G1, G2	Zone A
OR2	G1, G3	Zone A
OR3	G2, G4	Zone B
OR4	G3	Zone B
OR5	G4, G5	Zone C
OR6	G5, G6	Zone C

Receptor

Sensitive to...

OR1	Small, volatile molecules with citrus or minty notes
OR2	Cool or fresh molecules; also detects light sulfurs
OR3	Complex aromatics or citrus-aldehydes
OR4	Sweet-spicy compounds, especially multi-ring structures
OR5	Heavier molecules with sulfur or burnt odors
OR6	Intense or pungent volatiles, especially menthols and thiols

Odorants:

Developed by Julie Finnell Harris (University at Buffalo) for NRS coursework. Shared for Instructional discussion at FUN 2026.

- **Cineole:** eucalyptus; *volatile, mentholated, slightly bitter*
- **Lemonol:** citrus-like; *small, volatile, aldehydic*
- **Cinnaprene:** spicy-sweet; *complex, aromatic rings, warm*
- **Skunkamine:** sulfuric; *intense, pungent, thiol-based*
- **Minthex:** cool; *mentholated, fresh, bitter*

Using this information, complete the table. Note, a copy of this table is also found in the answer sheet. It has been added here for completeness and convenience. The answer to Cineole is provided as an example. You can work through this example and see if you get the same answer.

Odorant	Receptors activated	Paired Glomeruli	Cortical Zone
Cineole	OR1	G1, G2	Zone A
	OR6	G5, G6	Zone C
Lemonol			
Cinnaprene			
Skunkamine			
Minthex			

After completing the table consider these questions:

1. Look at the completed circuit. Which receptors and glomeruli were activated by more than one odorant? What might this suggest about how the brain encodes different smells using shared components?
2. Based on the overlap in activated receptors and glomeruli, which odorants might smell similar based on shared receptor/glomerulus activation? Are there shared chemical or perceptual features that support your prediction?
3. Imagine OR2 is damaged and can no longer respond to odorants. Which odorants would lose part of their input to the brain? How might this affect a person's ability to identify or distinguish these smells?